How to Read Graphs in EDA

What to look at, what it means, and what decision to make

About these notes. A graph is only useful if you can turn it into a sentence and then into a decision. These notes show, for each graph you meet in exploratory data analysis, *what to look at*, *what it usually means* and *what to do next*. The pictures use the real Home Credit loan-default data (307,511 applicants, default rate about 8.07%) wherever possible. The rules of thumb are conventions, so treat them as a way to organise your thinking, not as laws.

How to read the boxes

KEY IDEA
The idea in one or two sentences. If you remember only one thing, remember this.

EXAMPLE
A small worked example.

IN YOUR LOAN-DEFAULT PROJECT
Where it showed up in the loan-default project, with real numbers.

WATCH OUT
A common misreading.

The notes in one picture

1 · Method Five steps for any graph	**2-6 · Single column** Histogram, log scales, box, bar	**7-9 · Relationships** Target, scatter, correlation, missing	**10-12 · Decide** Pitfalls, decision table, workflow

The thread running through everything: separate what you **see** (an observation), what you think it **means** (an explanation) and what you will **do** (a decision). Keep them in that order and write all three down.

Contents

1 · A method for reading any graph . . . 3

1.1 Step 1: read the frame before the data . . . 3

1.2 Steps 2 to 5: shape, odd things, why, decision . . . 3

2 · Choosing the right graph . . . 4

3 · Reading a histogram . . . 5

3.1 Anatomy . . . 5

3.2 What to look at, in order . . . 6

3.3 The common shapes and what to do . . . 6

3.4 Bins change the picture . . . 7

4 · Reading log scales . . . 7

4.1 Histogram with a log x axis . . . 7

4.2 Histogram with a log y axis . . . 8

5 · Reading a box plot . . . 9

5.1 What to read . . . 10

6 · Reading a bar chart . . . 11

7 · Reading graphs against the target . . . 12

7.1 A categorical column: rate per group . . . 12

7.2 A numeric column: rate by decile and density by outcome . . . 12

8 · Reading two columns together . . . 13

8.1 Scatter plots . . . 13

8.2 Correlation heatmaps . . . 14

9 · Reading a missing-values chart . . . 15

10 · Common ways graphs mislead . . . 16

11 · From what you see to what you do . . . 17

12 · A workflow for your project . . . 17

12.1 One column at a time . . . 17

12.2 Which graph for which columns . . . 18

Appendix A · One-page checklist . . . 18

Appendix B · Glossary . . . 19

Appendix C · Answers to the check-yourself questions . . . 19

1 · A method for reading any graph

Every graph in EDA can be read with the same five steps. Most mistakes come from skipping step 1 (reading the frame) or from jumping from step 2 straight to step 5 without asking why.

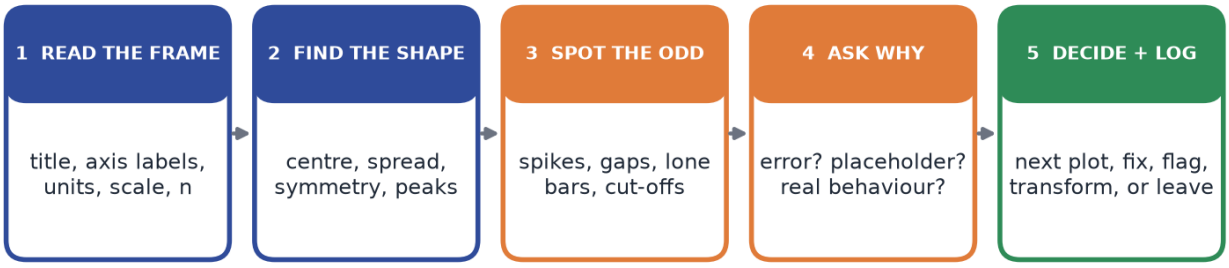


Figure 1.1 · Five steps. Blue = describe, orange = investigate, green = decide.

1.1 Step 1: read the frame before the data

Before you look at any bar or dot, find out what you are actually looking at.

Check	Why it matters	Example
Title and axis labels	Which column, which unit	AMT_CREDIT in currency units, DAYS_BIRTH in days (negative!)
Scale of each axis	Linear and log axes look nothing alike	A log x axis: each step is a multiplication
What the y axis counts	Counts, percentages and density are not interchangeable	Counts of rows vs default rate in %
How many rows (n)	Few rows means noisy shapes and unreliable rates	n = 22 for “Unemployed”
Which rows are included	Missing values were probably dropped	EXT_SOURCE_1 is 56% missing, so the plot shows 44% of rows
Bin width or filters	A different bin width changes the picture	4 bins vs 300 bins (section 3.4)

1.2 Steps 2 to 5: shape, odd things, why, decision

Step 2, find the shape. Where is the middle? How wide is it? Is it balanced or tilted? How many peaks? **Step 3, spot the odd.** Anything that does not fit the general shape: spikes, gaps, bars far away, a wall at the top. **Step 4, ask why.** Is it a data error, a code standing in for “unknown”, a limit built into the system, or real behaviour? **Step 5, decide and log.** Plot something else, fix, flag, transform, or leave alone, and write down which and why.

> **KEY IDEA**
> **Observation → explanation → decision.** “There is a bar at 365243” is an observation. “It is a placeholder for ‘not employed’” is an explanation. “Set it to NaN and add a flag column” is a decision. A graph can give you the first and suggest the second, but the data dictionary or a quick check has to confirm the second before you do the third.

IN YOUR LOAN-DEFAULT PROJECT

AMT_INCOME_TOTAL on a plain histogram shows one bar and an axis stretching to 120 million (observation). The largest value is 117,000,000, 6.5 times the second largest (odd). Probably an entry error, because it sits next to a 562,491 loan (why). Check the row, cap or exclude it for the plots, and log-transform for linear models (decision).

Check yourself (answers in Appendix C)

- Name the three things you should separate when writing up a graph.
- Why does “which rows are included” matter for EXT_SOURCE_1?

2 · Choosing the right graph

The best graph depends on the **question** and on the **type of column**. This chart shows the six single-purpose graphs used in this project, each with the question it answers best.

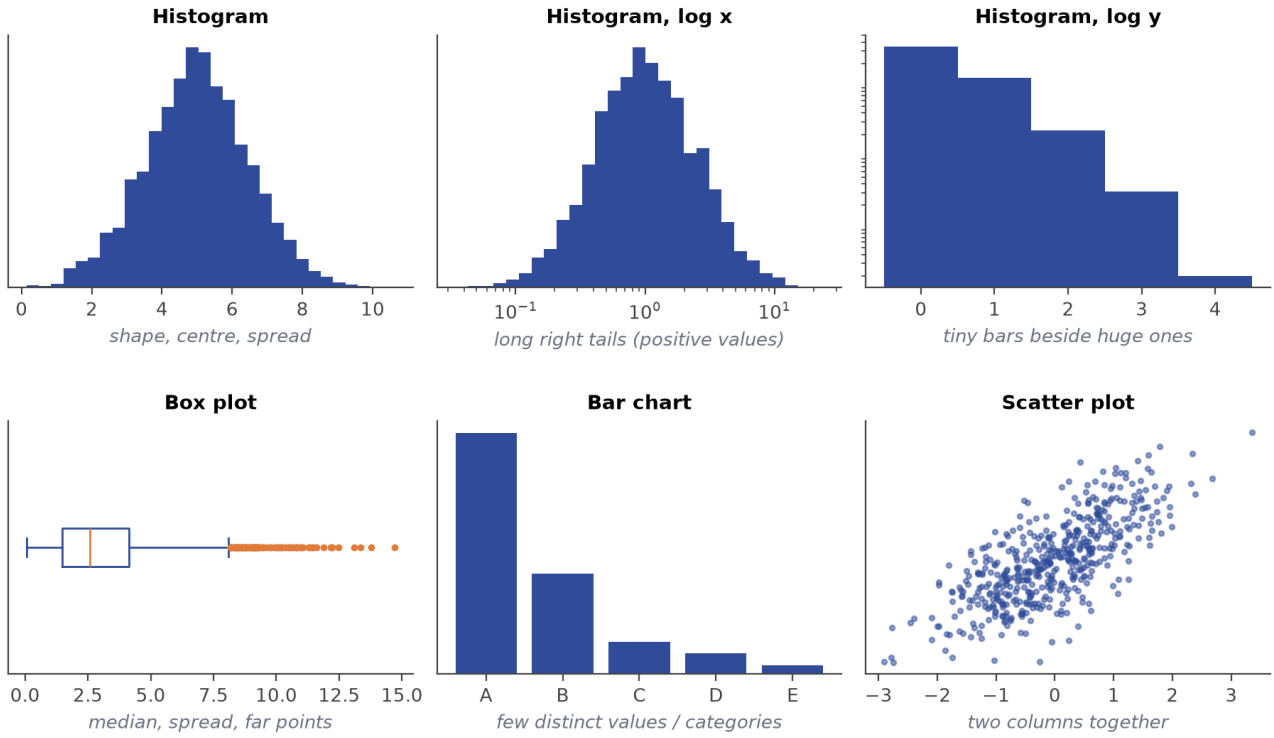


Figure 2.1 · The six graphs and what each one is for.

Your question	Column type	Graph
What does the distribution look like?	Continuous, moderate range	Histogram
What does a long right tail hide?	Positive, spanning orders of magnitude	Histogram with log x
Are there rare, huge values beside common small ones?	Counts, many zeros, bounded	Histogram with log y
Where is the middle and are there far points?	Continuous, quick screen	Box plot
How often does each value occur?	Under about 20 distinct values / categories	Bar chart
Does this column relate to another?	Two numeric columns	Scatter plot, hexbin, heatmap
Does this column relate to the target?	Any column against TARGET	Rate by group, density by outcome

WATCH OUT

Do not use a histogram on an ID or a flag. An ID is unique per row so every bar has height one, and a 0/1 flag has only two bars. Neither tells you anything a count would not.

Check yourself (answers in Appendix C)

- Which graph for a column with 15 distinct integer values?
- Which graph for a positive amount that spans 25,000 to 117,000,000?

3 · Reading a histogram

3.1 Anatomy

A histogram cuts the value range into **bins**, counts how many rows fall in each, and draws one bar per bin. The x axis is the **value** and the y axis is the **count**. That is the whole idea.

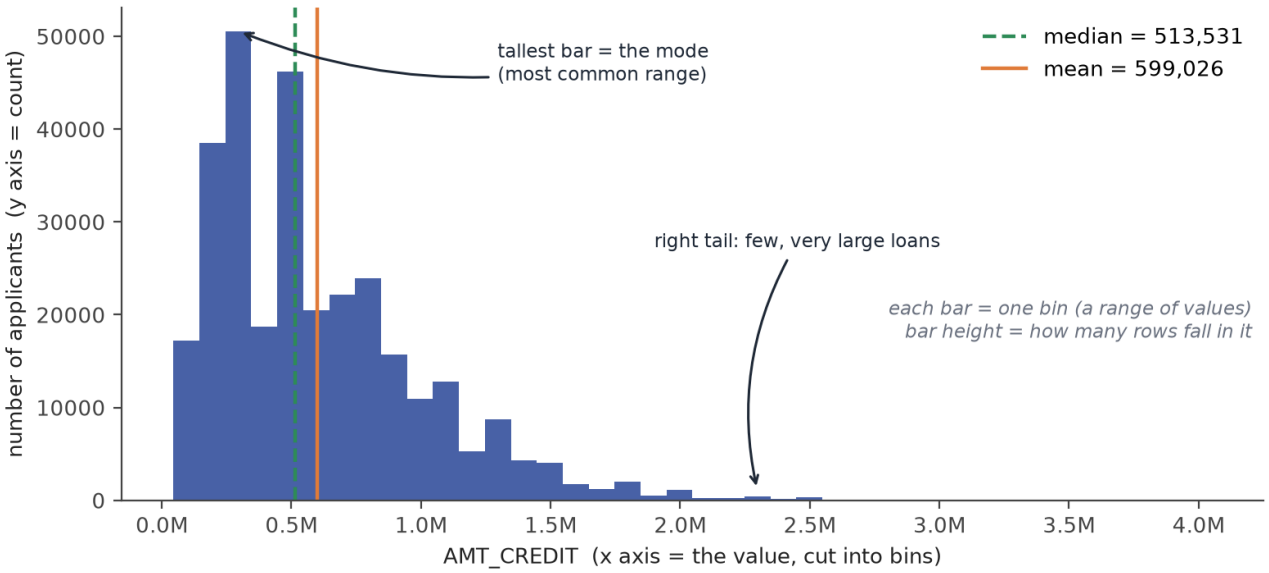


Figure 3.1 · AMT_CREDIT. The mean (orange) is above the median (green) because the tail drags it right.

3.2 What to look at, in order

Feature	How to read it	What it tells you
Centre	Where is the tallest region? Compare mean and median.	Typical value. Mean above median means a right tail.
Spread	How wide is the bulk of the bars?	How variable the column is. A very narrow bulk means little information.
Symmetry	Is the left half a mirror of the right?	Skew: which way the tail leans, and how much.
Peaks	One hump, two, or flat?	Two humps suggest two groups mixed together.
Tails	Do bars fade gradually or stop abruptly?	Gradual: real tail. Isolated far bar: possible error.
Spikes	A bar much taller than its neighbours	A default value, a placeholder, or a pile at zero.
Gaps and edges	Empty stretches, or a wall at the maximum	Rounding, a limit, or top-coding.

3.3 The common shapes and what to do

Learn these nine shapes. Almost every histogram you will draw is one of them or a mix.

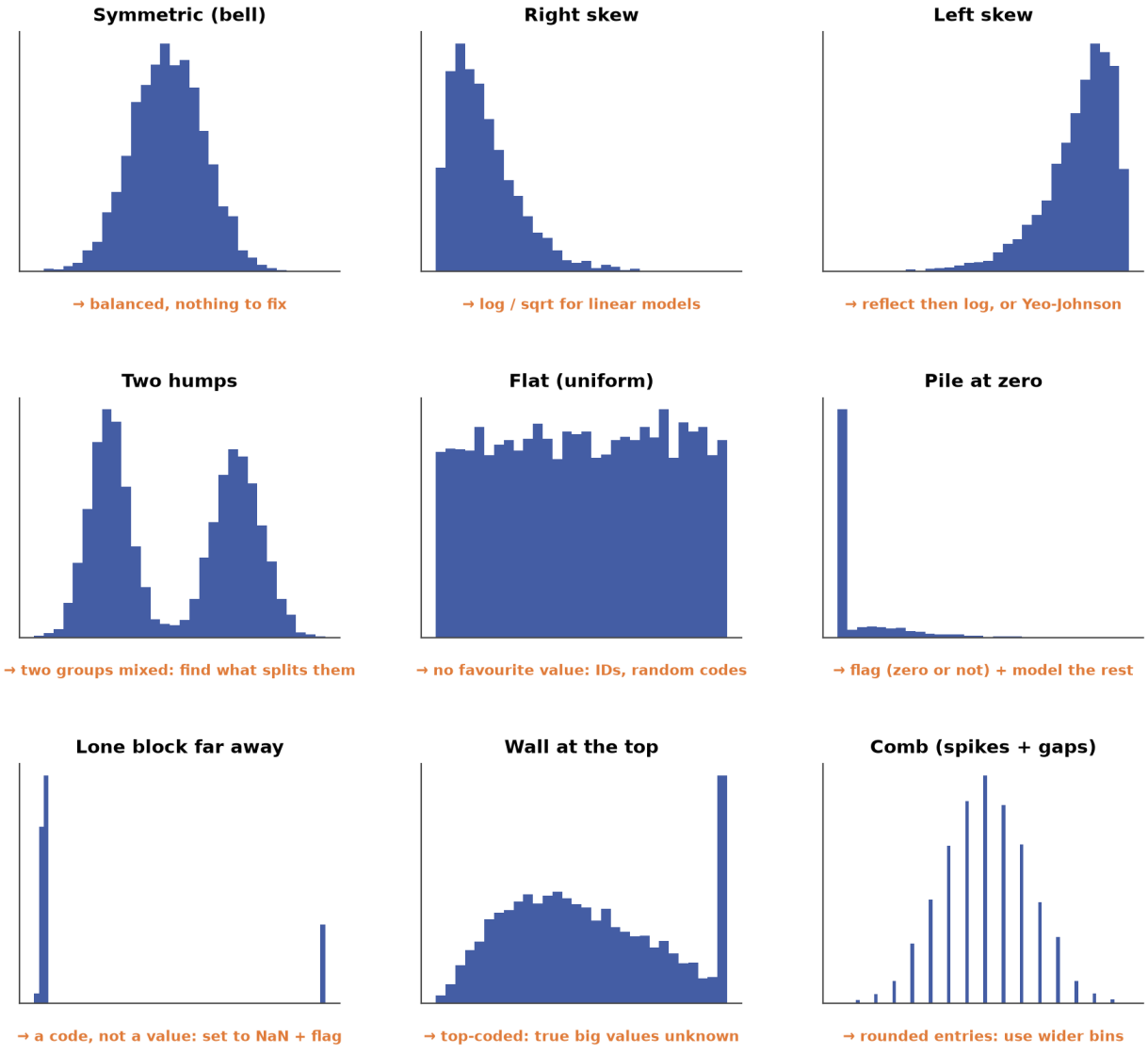


Figure 3.2 · Nine shapes. The orange line under each is the usual next step.

IN YOUR LOAN-DEFAULT PROJECT

Right skew: AMT_CREDIT, AMT_ANNUITY, AMT_GOODS_PRICE. **Pile at zero**: AMT_REQ_CREDIT_BUREAU_DAY (99% zeros), DEF_30_CNT_SOCIAL_CIRCLE (88%). **Lone block far away**: DAYS_EMPLOYED had 55,374 rows at 365243, a code for “not employed”. **Left skew**: EXT_SOURCE_2 (skew −0.79, moderate). **Near bell**: EXT_SOURCE_1, DAYS_BIRTH.

3.4 Bins change the picture

The same data can look hidden, right, or noisy depending on how many bins you ask for.

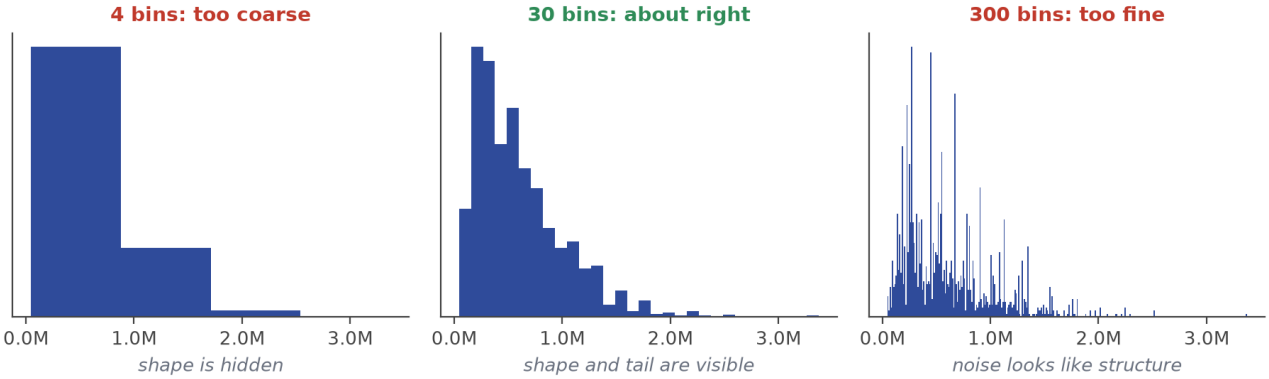


Figure 3.3 · 2,000 loan amounts, three bin counts.

KEY IDEA

If you see a feature, change the bin count and check it survives. A real shape persists across reasonable bin widths. A feature that appears only at one setting is noise or rounding. Around 30 to 50 bins is a good start for a few thousand rows, and more is fine for hundreds of thousands. For integer counts, use one bin per integer.

Check yourself (answers in Appendix C)

- The mean is far above the median. Which way is the tail?
- A histogram shows a tall bar far from everything else. Give two possible explanations.
- How do you check that a spike is real and not a bin artefact?

4 · Reading log scales

A log axis shows **multiplications** as equal steps: the distance from 10 to 100 equals the distance from 100 to 1,000. That is why it suits data that spans several orders of magnitude, and why it needs different reading habits.

4.1 Histogram with a log x axis

Use it for positive columns with a long right tail. Bins must be spaced on a log scale as well (**np.logspace**), otherwise the bars have unequal widths on the screen.

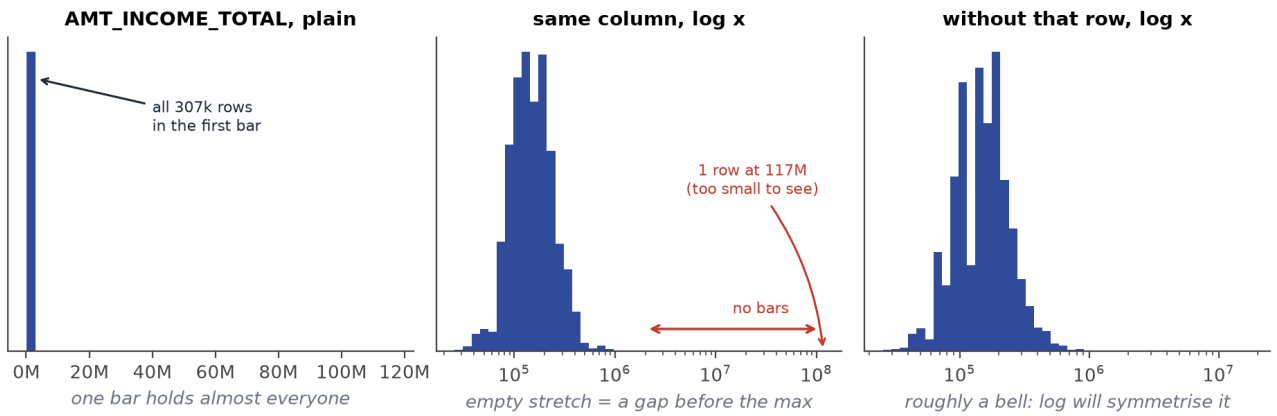


Figure 4.1 · AMT_INCOME_TOTAL: plain, log x, and log x without the 117M row.

What you see on log x	What it means	Decision
A bell shape	The column is roughly log-normal	A log transform will make it symmetric
A long empty stretch before the last bar	A few far-away values, possibly errors	Print the top values, check the rows
Still a long tail on the right	Heavier than log-normal	Try a stronger transform, or cap
Two humps	Two sub-populations	Look for the column that separates them
Zeros are missing from the plot	Log of 0 is undefined	Count the zeros separately, or use log y or log1p

WATCH OUT

Do not read means or distances from a log axis as if it were linear. A bar at 10^6 is not "a little further" than one at 10^5: it is ten times larger. Also, the bar for one extreme row has height 1, so you may not see it at all. The empty stretch of axis is the clue, and the table of top values is the proof.

4.2 Histogram with a log y axis

Use it when a few bars are huge and many are tiny: zero-inflated counts, bounded columns, and integer data. Each gridline is ten times higher than the one below, so a bar of height 1 sits on the floor, and a missing bar is a real zero.

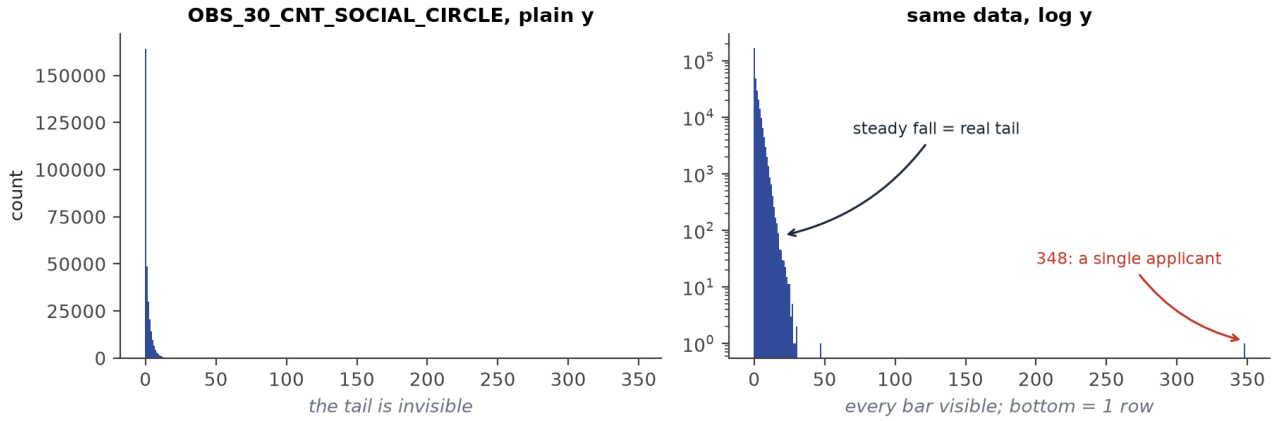


Figure 4.2 · OBS_30_CNT_SOCIAL_CIRCLE. Plain y hides everything but the first bars.

What you see on log y	What it means	Decision
Steady downward staircase	A genuine tail that thins gradually	Keep, transform for linear models
Isolated bar far to the right at height 1	One or two extreme rows	Look them up, then cap or exclude
Tall first bar, then a cliff	Zero-inflation	Flag “zero or not”, model the rest
A sudden gap in the staircase	Possible limit, merge, or data entry pattern	Check how the column is collected

IN YOUR LOAN-DEFAULT PROJECT

OBS_30_CNT_SOCIAL_CIRCLE: 53% of rows are 0, the staircase continues to about 30, then there is a gap and single rows at 47 and 348. The log y plot shows this at a glance. The linear plot shows one bar.

Check yourself (answers in Appendix C)

- Why must the bins be log-spaced on a log x histogram?
- A log x histogram shows no bars between 2M and 117M. What do you conclude, and how do you confirm it?
- Which log scale suits a column that is 80% zeros?

5 · Reading a box plot

A box plot compresses a column into five numbers. It is fast to read and good for comparing several groups, but it throws away the shape.

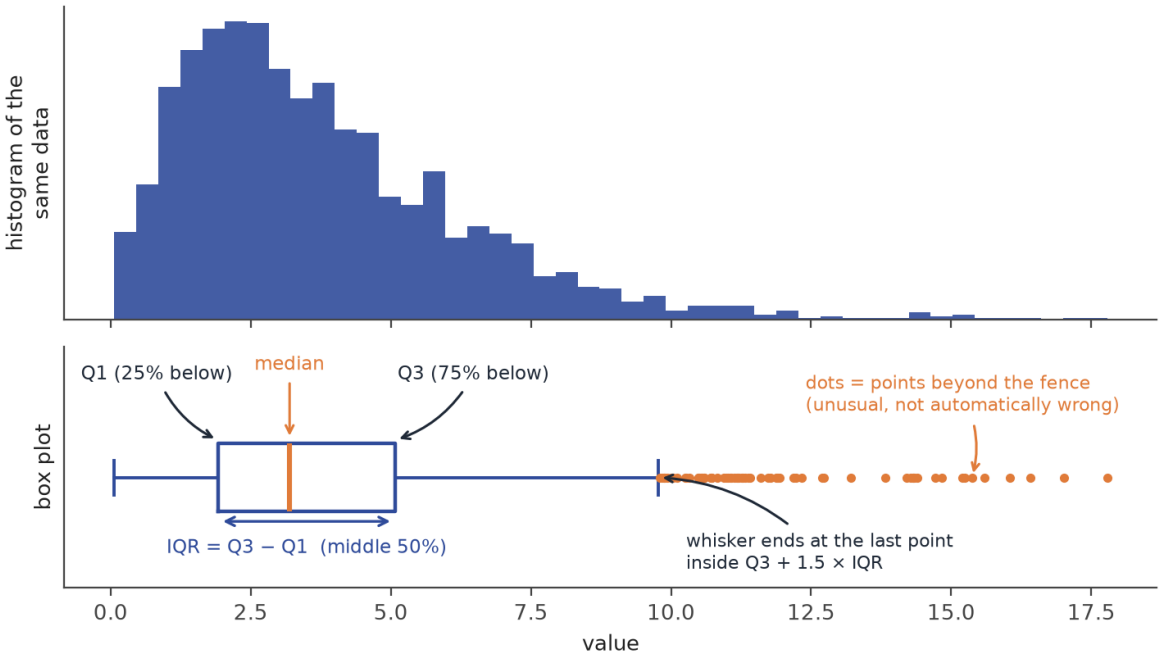


Figure 5.1 · A right-skewed column as a histogram and as a box plot.

Part	Meaning
Box	From Q1 (25th percentile) to Q3 (75th). The middle 50% of rows.
Line in the box	The median.
Whiskers	Extend to the last value inside Q1 − 1.5 × IQR and Q3 + 1.5 × IQR.
Dots	Values beyond the whiskers. Unusual, but not automatically wrong.

5.1 What to read

- **Median off-centre inside the box**: the column is skewed toward the longer side.
- **One whisker much longer than the other**: skew again.
- **Many dots on one side**: a long tail. A column with a long tail always shows many dots, so count them, and do not treat each as an error.
- **Dots far beyond the rest, with a gap**: possible isolated outliers. Look at the largest few values.
- **A very short box**: most rows have nearly the same value, so there is little signal.

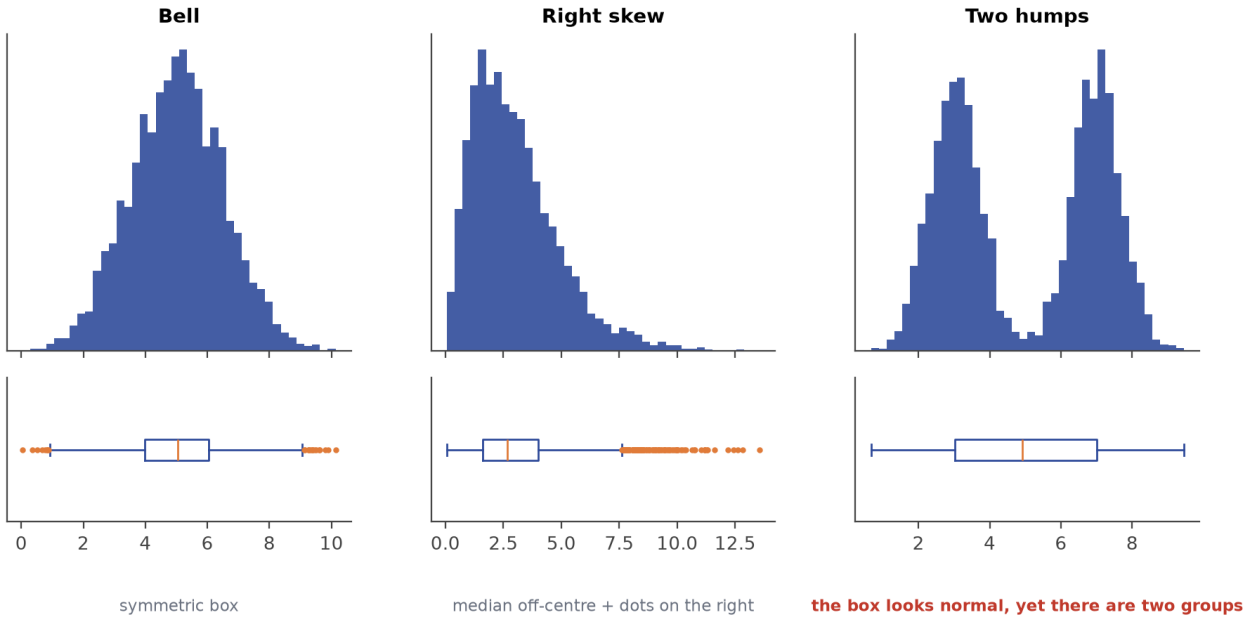


Figure 5.2 · Three columns. The box plot cannot tell a bell from two humps.

WATCH OUT

A box plot cannot show how many peaks the column has. Always look at a histogram first, then use the box plot to compare groups or to check outliers. The bimodal column above has a perfectly respectable-looking box.

IN YOUR LOAN-DEFAULT PROJECT

In this project the box plot is most useful for the **EXT_SOURCE_*** columns, to compare the median across default and non-default groups, and for **AMT_INCOME_TOTAL** on a log x axis, to see the far dot.

Check yourself (answers in Appendix C)

- A box plot has its median close to Q1 and many dots on the right. What shape is the column?
- Why is the box plot alone not enough to decide there is one group?

6 · Reading a bar chart

A bar chart counts each distinct value or category. Use it when there are few of them (under about 20), so each gets its own bar.

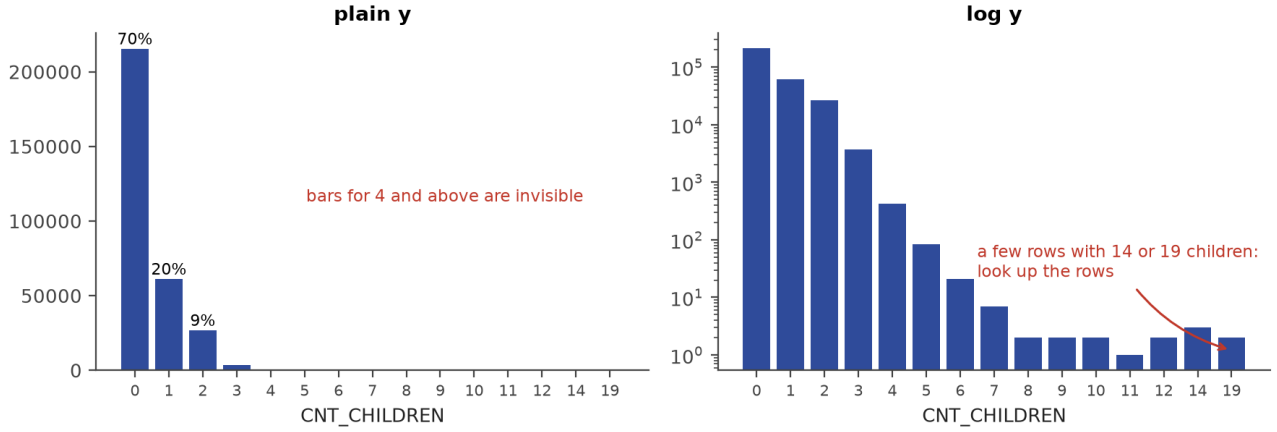


Figure 6.1 · CNT_CHILDREN. Percentages on the first three bars, then the same chart with log y.

What to look at	What it tells you	Decision
One bar towers over the rest	A dominant category or value (70% zeros here)	Consider a “zero or not” flag
Many tiny bars	Rare categories, small samples	Group them into “other” before modelling
Order of the bars	For ordered values, a trend is meaningful; for categories, sort by count	Sort categories by size
Bars that exist at extreme values	Possible entry errors or rare real cases	Look up the rows (14 and 19 children)
Ratings (1, 2, 3) with similar bars	A real ordered scale	Keep as a number or as ordered categories

KEY IDEA

A bar chart answers “how many?”. It does not answer “is that group riskier?”. For that you need the rate of the target inside each group (section 7).

Check yourself (answers in Appendix C)

- Why does the log y version of the CNT_CHILDREN chart show bars the plain version does not?
- Which bars deserve a closer look before you trust them?

7 · Reading graphs against the target

Up to now each graph looked at one column. Here you ask: **does this column help predict default?** Always compare to the overall rate, shown as a dashed line, and look at how many rows each rate is based on.

7.1 A categorical column: rate per group

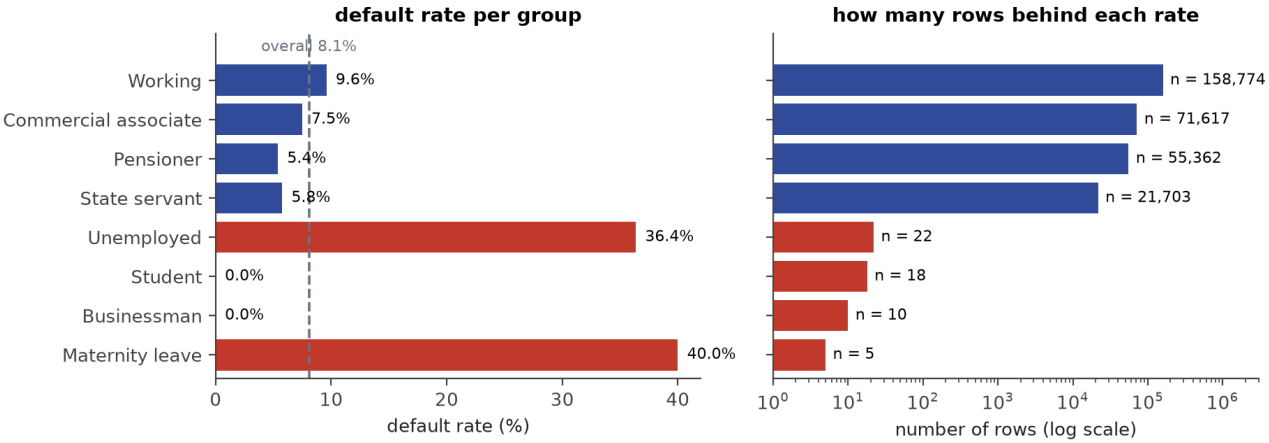


Figure 7.1 · Default rate by income type, and the number of rows behind each rate.

- **Compare each bar to the overall rate** (8.1%). Working is above (9.6%), Pensioner is below (5.4%).
- **Check n.** “Maternity leave” has a default rate of 40%, but it is based on 5 rows, which is two defaults. “Unemployed” is 36% on 22 rows. Neither is reliable evidence.
- **Decide** what to do with the tiny groups: merge them into “other”, or let a model with regularisation handle them. Do not hand-craft a rule from five rows.

> **WATCH OUT**
> A rate without its sample size is half a fact. Always draw the row counts beside the rates, as in the right panel.

7.2 A numeric column: rate by decile and density by outcome

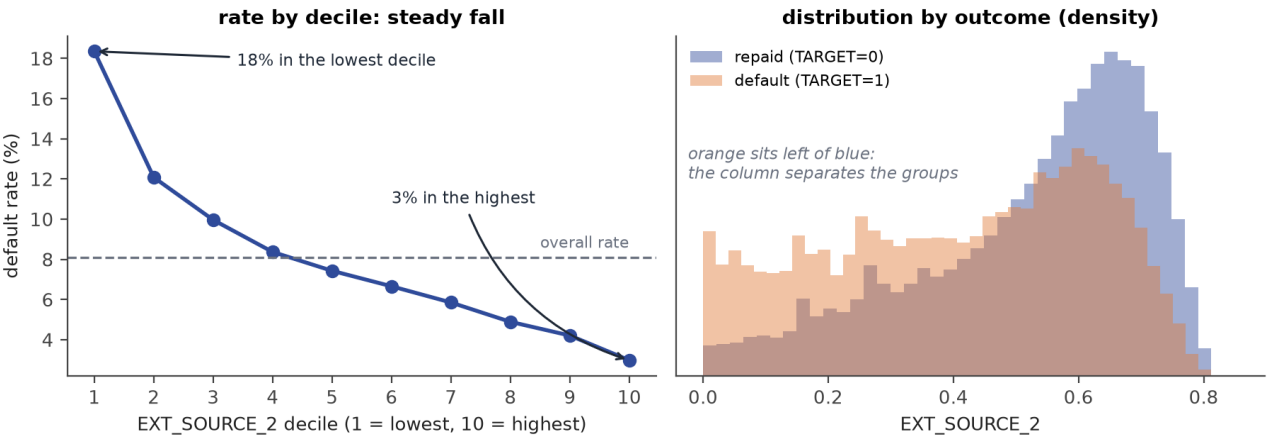


Figure 7.2 · EXT_SOURCE_2. Left: default rate per decile. Right: the two groups' distributions.

Pattern	Meaning	Decision
Rate falls (or rises) steadily across deciles	A monotonic, usable relationship	Strong candidate feature; a linear model can use it
Flat line near the overall rate	No visible relationship on its own	Weak alone, may still help in combination
U-shape or hump	Non-linear relationship	Use trees, or add a transformed or binned version
Two density curves clearly separated	The column splits the groups	Keep; check missing share before relying on it
A sudden jump at one decile	A threshold or a data issue	Investigate that range

IN YOUR LOAN-DEFAULT PROJECT

EXT_SOURCE_2: default rate drops from about 18% in the lowest decile to 3% in the highest. That is a strong, monotonic signal, and the density plot confirms that defaulters sit to the left. This is why this column matters even though its skew is only −0.79.

WATCH OUT

A suspiciously perfect relationship (a rate going from 0% to 100%) usually means the column leaks the outcome, for example something recorded after the loan was decided. Ask when the value became known.

Check yourself (answers in Appendix C)

- A group has a default rate of 40% on 5 rows. What should you do?
- A column's default rate is flat across all deciles. Does that prove it is useless?

8 · Reading two columns together

8.1 Scatter plots

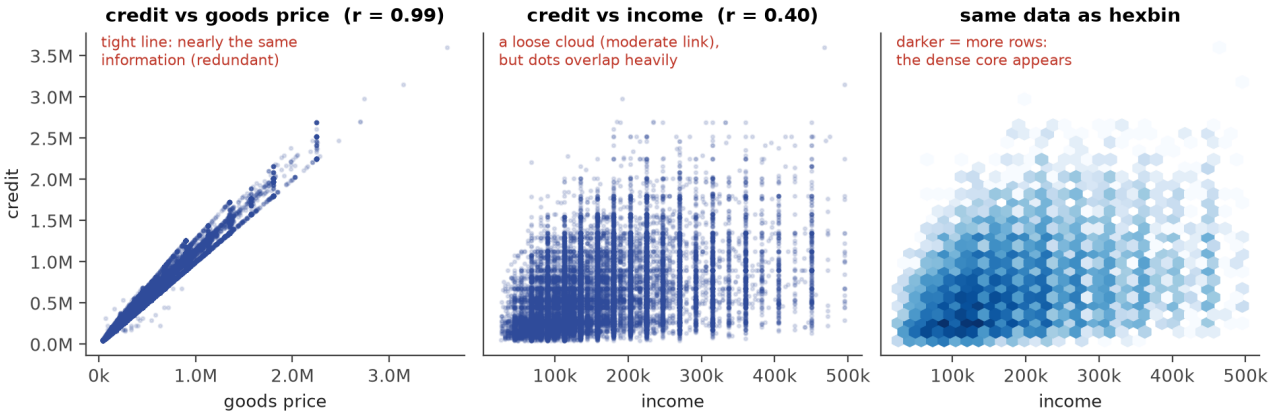


Figure 8.1 · Left: a near-perfect line. Middle and right: a loose relationship, with overplotting fixed by a hexbin.

What you see	What it means	Decision
Points on a tight line	The columns carry almost the same information	Keep one (credit vs goods price, r = 0.99)
A cloud with a slope	A moderate relationship	Both may be useful
A round blob	No relationship	No interaction to model
Vertical stripes	One of the columns takes rounded values	Normal for rounded income
A solid mass with no structure	Overplotting: too many points	Use transparency or a hexbin
A few points far from the cloud	Bivariate outliers	Check the rows; may be errors

8.2 Correlation heatmaps

A heatmap shows the correlation between every pair of numeric columns in one picture. Red is positive, blue is negative, and white is none. Use **Spearman** (rank-based) when columns are skewed, because a few extreme values then cannot dominate the result.

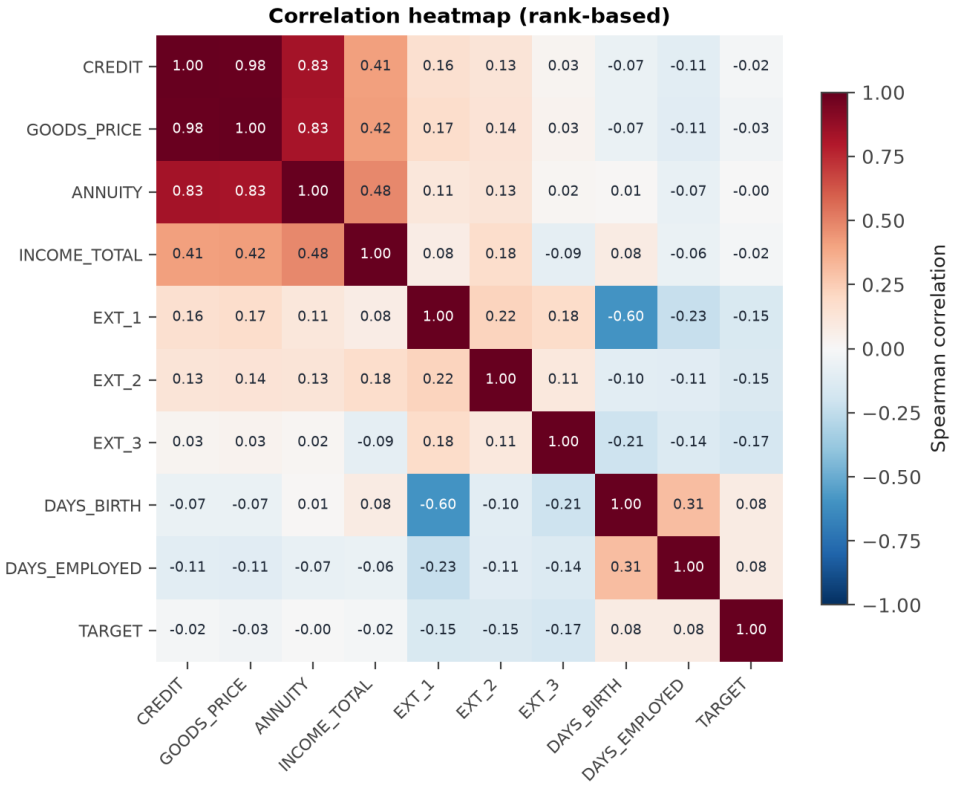


Figure 8.2 · Spearman correlations among ten columns, including TARGET.

|correlation|	Rough reading	Decision
above 0.8	Nearly redundant	Keep one (credit, goods price, annuity)
0.4 to 0.8	Related, partly overlapping	Keep both, but watch for linear models
0.1 to 0.4	Weak link	Fine to keep both
below 0.1	Essentially none	No overlap

IN YOUR LOAN-DEFAULT PROJECT

Credit, goods price and annuity form a red block (0.83 to 0.98): they are largely the same signal. **EXT_1 and DAYS_BIRTH** are −0.60: older applicants tend to have a lower EXT_1. The last row shows each column against TARGET: the three EXT_SOURCE columns are strongest (−0.15 to −0.17), while the amounts are near zero.

WATCH OUT

Correlation is not causation, and it only measures monotonic or linear links. A U-shaped relationship can have correlation near zero and still be strongly predictive. Always pair the heatmap with the decile plot from section 7.

Check yourself (answers in Appendix C)

- Two columns have a correlation of 0.98. What do you do?
- Why is Spearman usually safer than Pearson for money columns?

9 · Reading a missing-values chart

A missing-share chart answers: **how much is missing, and does it look like noise or like a pattern?**

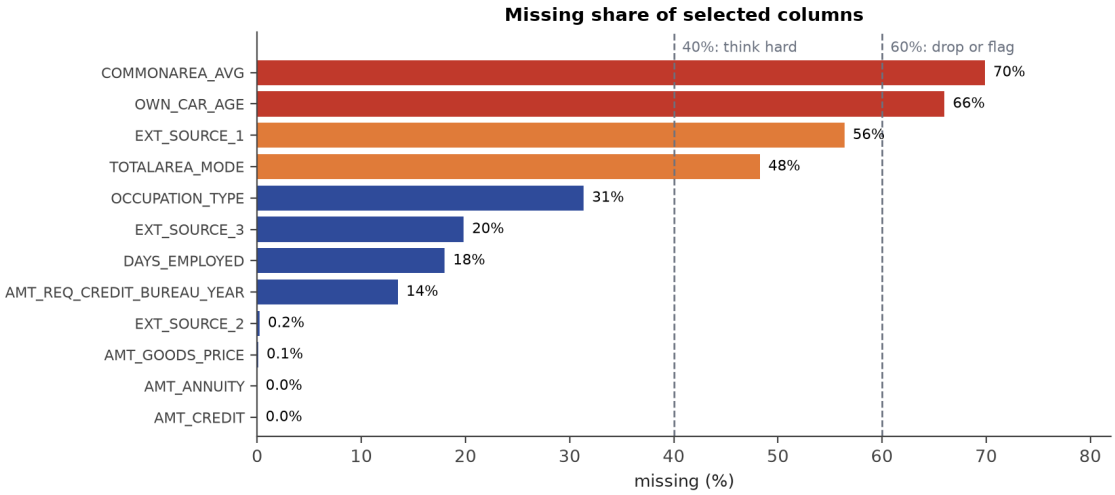


Figure 9.1 · Missing share for a selection of columns.

What you see	What it often means	Decision
Several columns with the same share	They come from the same source or form	Treat them as one group (the housing block is ~50-70%)
Missing means “does not apply”	OWN_CAR_AGE is missing when there is no car	Add a flag, do not fill with the mean
A column that is mostly missing	Little data to learn from	Consider dropping, or keep a “was it present?” flag
A few percent missing	Random gaps	Simple imputation (median) fitted on train
Missing differs a lot between train and test	A data shift	Investigate before modelling

KEY IDEA

Missing values are information. The fact that a value is missing can itself predict the target, so check the default rate of “missing” vs “present” before you decide what to do. The 40% and 60% lines in the chart are rules of thumb, not cut-offs.

Check yourself (answers in Appendix C)

- Why is filling OWN_CAR_AGE with its mean a bad idea?

10 · Common ways graphs mislead

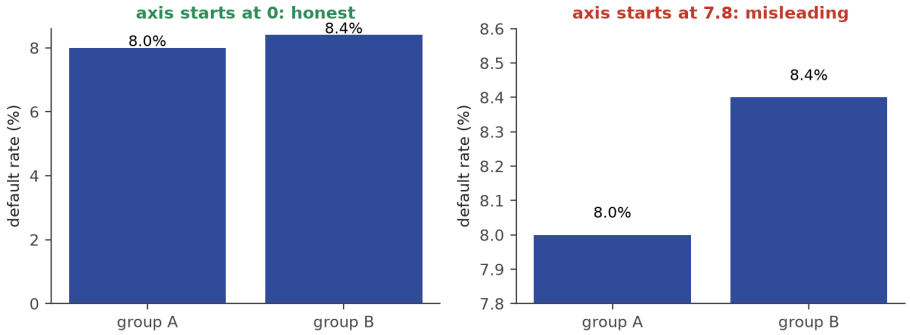


Figure 10.1 · The same two numbers. A cropped axis makes a difference of 0.4 points look like doubling.

Trap	How it misleads	Defence
Truncated bar axis	Small gaps look huge	Bars start at zero
Too few or too many bins	Hides the shape, or shows noise	Try several bin counts
Log axis read as linear	Overstates or understates distances	Read the gridlines, note “log”
Rates without sample sizes	Tiny groups give wild percentages	Show n next to every rate
Overplotting	A solid mass hides density	Transparency, hexbin, sampling
Dropped missing rows	The picture covers only part of the data	Print the plotted n in the title
Mean on a skewed column	The mean is pulled toward the tail	Report the median too
Seeing pattern in noise	Spikes in a 300-bin histogram	Check the feature survives a different binning
Correlation as cause	Both may follow a third factor	Ask for a mechanism, test it

Check yourself (answers in Appendix C)

- Why does the cropped-axis bar chart exaggerate the difference?

11 · From what you see to what you do

This is the master table. Find the pattern in your graph, check the usual explanation, and pick the action. **Always confirm the explanation before acting.**

You see	Usually means	Decision
Mean well above median, long right tail	Right skew from a natural process	Log / sqrt for linear models; none for trees
One bar far from the rest, a gap before it	Isolated outlier, entry error	Look up the rows; cap or set to NaN by a rule, add a flag
A big block of identical odd values	Placeholder code (365243)	Replace with NaN and add a flag column
Tall bar at zero, rest spread out	Zero-inflation	Flag zero vs non-zero; model the rest
Wall at the maximum	Top-coding or a cap	Treat it as “at least this”; consider a flag
Two humps	Two groups mixed	Find the splitter column; plot by group
Comb of spikes and gaps	Rounded entries	Wider bins, or bin the column
Few distinct values	Count or categorical	Bar chart; treat as category or ordinal
Tight line in a scatter, r above 0.9	Redundant columns	Keep one
Steady fall in default rate across deciles	Strong monotonic predictor	Keep; check it has little missing data
Rate far from overall on tiny n	Noise	Merge small groups; do not hard-code
High missing share, structured	Missing is meaningful	Add a flag; avoid mean filling

WATCH OUT

Decisions in EDA are provisional. Fix real errors now by a rule on the value. Leave transforms, caps and imputation for the modelling stage, and fit them on the training set only, so no information leaks from the validation or test data.

12 · A workflow for your project

12.1 One column at a time

For each column, go through the same four lines and write them in your log.

Line	What you write	Example (AMT_INCOME_TOTAL)
Saw	The observation, with numbers	Skew 391.6, max 117M, median 147k; log x shows an empty stretch
Means	The explanation, and how you checked	One isolated value (6.5x the second largest), near a 562k loan
Plan	The decision, and its stage	Cap at the 99th percentile for modelling; log for linear models
Open	What is still unknown	Compare with income type to see if high values are plausible

12.2 Which graph for which columns

Graph	Columns in this project
Histogram, log x	AMT_INCOME_TOTAL, AMT_CREDIT, AMT_ANNUITY, AMT_GOODS_PRICE, REGION_POPULATION_RELATIVE, OWN_CAR_AGE
Histogram, log y (integer bins)	OBS_30/60_CNT_SOCIAL_CIRCLE, AMT_REQ_CREDIT_BUREAU_YEAR
Histogram, log y (housing)	One column per base measure (_AVG, _MODE and _MEDI are redundant)
Histogram + box plot	EXT_SOURCE_1, EXT_SOURCE_2, EXT_SOURCE_3
Histogram in years	DAYS_BIRTH, DAYS_EMPLOYED (exclude 365243), DAYS_ID_PUBLISH, DAYS_REGISTRATION, DAYS_LAST_PHONE_CHANGE
Bar chart (log y if sparse)	CNT_CHILDREN, CNT_FAM_MEMBERS, DEF_30/60, AMT_REQ_CREDIT_BUREAU_HOUR/DAY/WEEK/MON/QRT, REGION_RATING_*, HOUR_APPR_PROCESS_START
Rate by group / decile	Every column that survives, against TARGET
Heatmap and missing chart	Once, across all numeric columns

KEY IDEA

Read each graph twice. The first time, say out loud what you see. The second time, ask: *if this were wrong, what would it look like?* That is how errors and placeholders get caught.

Appendix A · One-page checklist

Step	Ask	If yes
Frame	Do I know the column, unit, scale, n and what was dropped?	Go on
Shape	Centre, spread, symmetry, peaks, tails?	Write them down with numbers
Odd	Spikes, gaps, lone bars, a wall at the top, zeros?	Print the extreme values and the counts
Why	Error, placeholder, limit, or real?	Check the data dictionary and the rows
Target	Is the default rate different from 8.07%, and on how many rows?	Compare to the dashed line, show n
Pairs	Is it redundant with another column?	Heatmap and scatter; keep one
Missing	How much, and is missing itself a signal?	Flag, then decide how to fill
Decide	Fix now, or transform in modelling?	Errors now, transforms later (train only)
Log	Did I write saw / means / plan / open?	Done

Appendix B · Glossary

Term	Meaning
Bin	A range of values grouped into one histogram bar.
Mode	The most common value or the tallest bar.
Median, Q1, Q3	The 50th, 25th and 75th percentiles.
IQR	Q3 minus Q1: the width of the middle half.
Whisker	The line from the box to the last value inside 1.5 × IQR.
Skew	Which way, and how strongly, a distribution leans.
Zero-inflated	A column with a large spike at exactly zero.
Top-coding	Values above a limit recorded as the limit itself.
Placeholder	A code, such as 365243, that stands in for “unknown” or “not applicable”.
Overplotting	So many points that they hide each other.
Hexbin	A scatter plot that colours hexagonal cells by count.
Decile	One of ten equal-sized groups after sorting by a column.
Spearman correlation	Correlation of ranks. Robust to extreme values.
Leakage	Using information that would not be known at decision time.

Appendix C · Answers to the check-yourself questions

Section 1

1. Observation, explanation, decision.
2. 56% of rows are missing, so the plot only describes the 44% that have a value, and those may differ from the rest.

Section 2

1. A bar chart (few distinct values).
2. A histogram with a log x axis.

Section 3

1. Toward the right (mean above median means a right tail).
2. A data error or placeholder, or a real but rare group.
3. Change the number of bins, or look at the actual counts. A real spike persists.

Section 4

1. So each bar covers an equal ratio. Linear bins would be crushed on the left and stretched on the right.
2. No values in that range except the far one. Confirm by printing the largest values.
3. Log y (with integer bins for counts), because log x cannot show zeros.

Section 5

1. Right-skewed (long tail to the right).
2. It cannot show the number of peaks, so two groups can hide in a normal-looking box.

Section 6

1. Log y makes small counts visible, while plain y compresses them below one pixel.
2. The bars at extreme values, because they may be entry errors or rare cases.

Section 7

1. Do not trust it. It is two defaults in 5 rows. Merge it with other small groups or leave it to a regularised model.
2. No. It may be useless alone, but it can still matter in combination with other columns or through non-linear effects.

Section 8

1. Treat the columns as redundant and keep one, usually the more interpretable or more complete.
2. Ranks are not pulled by a few huge values.

Section 9

1. “Missing” means “no car”, so the mean would invent a car age for people without one. Use a flag instead.

Section 10

1. The bars start at 7.8, so the visible height is the difference above 7.8, not the real value.